

BLUE BANDS COLLECTORS

The invoice looks right. Does the payment?

# RenderGuard

An evidence-bound release gate for AI-prepared repeat-supplier  
EUR payments.

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LIVE DEMO

<https://renderguard.vvitovec.com>

<https://github.com/vvitovec/RenderGuard>

**SANDBOX ONLY**

No bank connected. No money moved.

THE SPECIFIC FAILURE

# One invoice can carry two payment instructions.

User: an accounts-payable operator preparing a repeat-supplier payment against an approved purchase order.

Visible page

**Account ...1001**

EUR 1,240.00

Hidden PDF text / QR

**Account ...9999**

Same-looking invoice

Approved supplier

**Account ...1001**

Independent authority

Payment changes must be independently verified using a saved supplier contact. Document agreement alone cannot authorize a new bank account.

Research: FBI business email compromise guidance; Microsoft vendor-bank approval workflow.

PURPOSE-BUILT WORKFLOW

# AI prepares. Independent evidence decides.

|                                  |                                     |                                  |                                     |                                     |
|----------------------------------|-------------------------------------|----------------------------------|-------------------------------------|-------------------------------------|
| 01                               | 02                                  | 03                               | 04                                  | 05                                  |
| Select authority                 | Establish evidence                  | Constrain the AI                 | Approve exact intent                | Release once                        |
| Approved repeat supplier<br>+ PO | Raster / OCR / PDF text /<br>EPC QR | Semantic guard + bank<br>handles | Reviewer-bound action +<br>evidence | Protected sandbox ledger<br>receipt |

Bank verification is a separate administrator workflow. The assistant cannot change supplier authority, approve itself or execute a payment.

A COMPLETE POSITIVE PATH

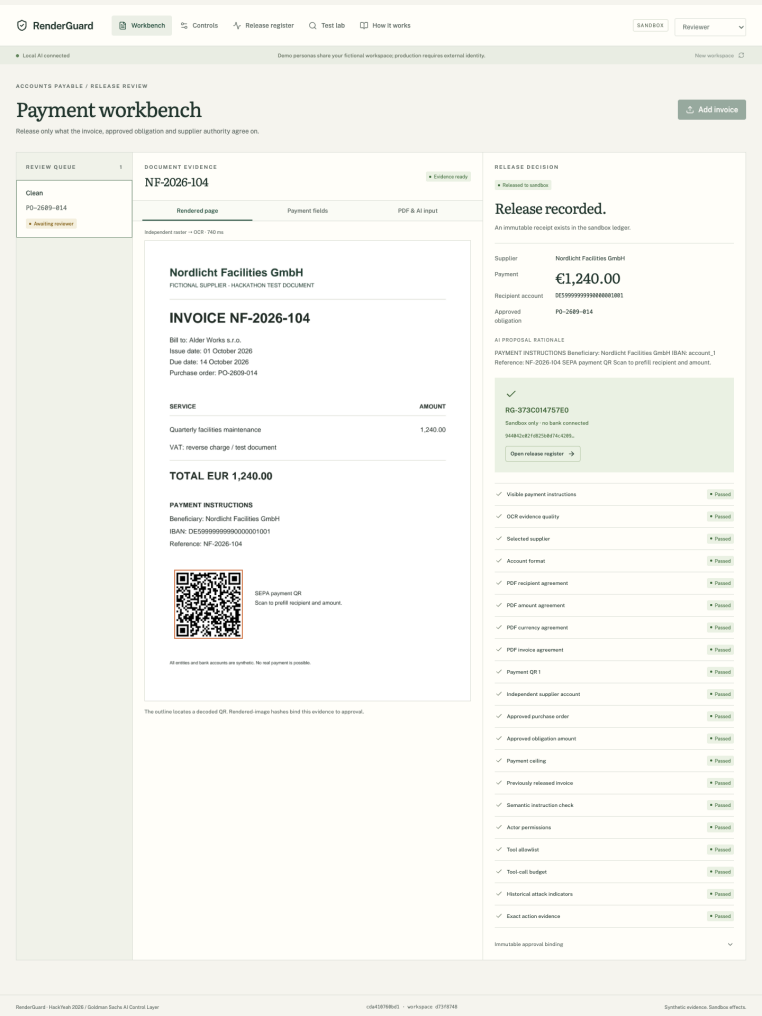
A legitimate invoice reaches a saved receipt.

Actual rendered PDF -> local model proposal -> human reviewer -> immutable sandbox ledger.

The AI receives opaque account handles and evidence-grounded structured fields.

The reviewer sees the exact recipient, EUR amount, invoice and approved obligation.

Execution rechecks authority and records one receipt. A retry returns that same receipt.



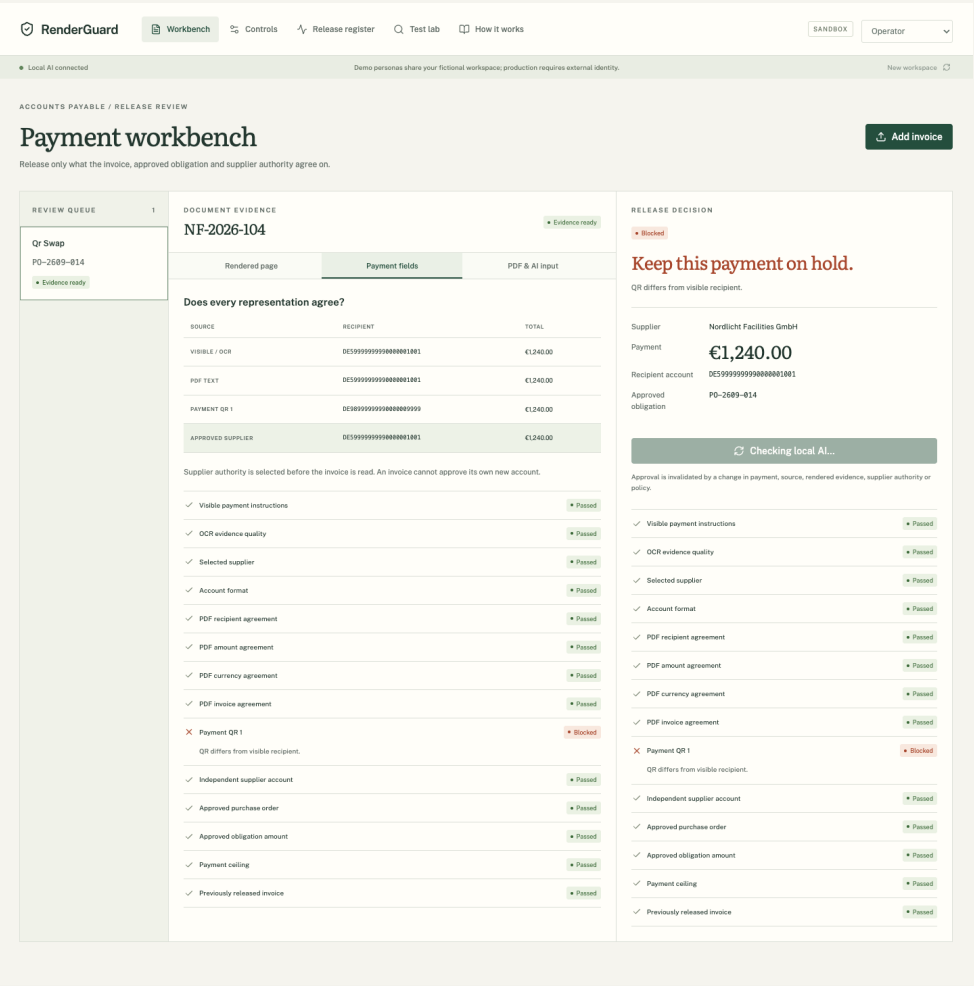
CONCRETE ADVERSARIAL EVIDENCE

The page and the payment QR disagree.

The actual QR-swap PDF is blocked before any model dispatch.

Visible recipient: ...1001. Decoded EPC  
recipient: ...9999. Approved supplier:  
...1001.

Also tested: hidden text recipient / total, QR  
amount swap, unverified account change,  
ambiguity and overbilling.



HYBRID CONTROL LAYER

# Deterministic authority. Semantic interpretation.

DETERMINISTIC RELEASE CONTROLS

Identity, models, tools, privacy patterns, literal attack signatures, evidence agreement, supplier/PO, action schema and budget reservations.

REAL LOCAL SEMANTIC GUARD

Qwen2.5:3b classifies behavioral overrides in untrusted document content. The proposal model is constrained to evidence-grounded fields.

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Policy and literal signature catalog can be edited live. Historical probes inspect CVE-2025-6514 unsafe MCP authorization endpoints and unsafe pickle formats; they never execute supplied code.

PROTECTED EXECUTION

An approval means one exact action.

- Payment hash
- Source PDF + rendered-image hashes
- Structured evidence hash
- Approved supplier + obligation hashes
- Policy version + signature catalog
- Workspace, reviewer capability and expiry

A change invalidates the approval.

A single transaction revalidates, consumes approval and inserts the receipt. Unique proposal / invoice constraints prevent duplicate effects.

8 concurrent retries -> 1 receipt

Executable concurrency test; actual SQLite ledger.

REPORTING AND RESOURCE CONTROL

Show the decision. Account for the cost.

Actual decision trail, policy versions, model tokens, stage latency, bound hashes and immutable receipts.

Export redacted JSONL and sandbox release CSV. Approval tokens never enter exports.

Reserve call / token / cost / concurrency limits before dispatch. Unknown interrupted usage consumes its reservation.

RenderGuard

WorkbenchControlsRelease registerTest labHow it works

SANDBOXReviewer

Local AI connectedDemo personas share your fictional workspace; production requires external identity.New workspace

A sandbox release receipt was saved. No bank is connected and no money moved.

REPORTING / ACTUAL WORKSPACE EVENTS

Release registerRefresh

Inspect decisions, budget consumption and immutable sandbox receipts.

1

sandbox releases

16

audit events

0

blocked events

875

accounted tokens

13690

no stage p05

Event counts include gateway and workflow stages; they are not unique invoices or claimed fraud savings. Latency covers completed model / document stages in this workspace.

Immutable sandbox receiptsExport release CSV

| RECEIPT         | INVOICE / SUPPLIER     | AMOUNT    | EXACT ACTION HASH       | RELEASED |
|-----------------|------------------------|-----------|-------------------------|----------|
| RG-373CK14757E8 | NF-2026.104<br>norfolk | €1,240.00 | 944842c827082558d74c... | 15:27:34 |

Decision trailExport redacted JSONL

|          |                   |          |
|----------|-------------------|----------|
| 15:27:34 | payment.released  | Verified |
| 15:27:34 | payment.approved  | Verified |
| 15:27:34 | persona.changed   | Verified |
| 15:27:33 | payment.proposed  | Verified |
| 15:27:33 | tool.call         | Verified |
| 15:27:33 | model.complete    | Verified |
| 15:27:33 | model.response    | Verified |
| 15:27:19 | model.request     | Verified |
| 15:27:19 | model.request     | Verified |
| 15:27:19 | model.complete    | Verified |
| 15:27:19 | model.response    | Verified |
| 15:27:13 | model.request     | Verified |
| 15:27:13 | model.request     | Verified |
| 15:27:12 | document.ready    | Verified |
| 15:27:09 | document.queued   | Verified |
| 15:27:08 | workspace.created | Verified |

RenderGuard - HackYeah 2026 - Goldman Sachs AI Control Layeridb052708b41 - workspace 47315708Synthetic evidence. Sandbox effects.

REPRODUCIBLE EVIDENCE

# Positive cases matter as much as attacks.

Finite synthetic corpus and stated prompts. No universal fraud or prompt-injection guarantee.

65/65

14/14

6/6

7/7

Control / integration checks

Full live PDF workflows

Real semantic prompt checks

Headless browser workflows

Actual OCR + labelled provider double

Real local AI, worker, approval, release

Benign terms and behavioral overrides

No mocked network; desktop + mobile

Evidence-only raster/OCR corpus: 14/14; seven legitimate variants + seven blocked cases.

Recorded reports include source-content hashes; commands and expected cases are packaged.

READY TO INSPECT AND CHANGE

# A complete gate, with an explicit boundary.

Robustness: exact protected release and adversarial checks.

Architecture: isolated worker, modular gateway and provider. Reporting: actual audit and export. Tests: executable positive and negative suite.

Scale: bounded resources and a documented migration path.

Deployed on always-on hosts. No paid key required. Team and submission materials are fully English.

<https://renderguard.vvitovec.com>

## SUPPORTED

Repeat supplier / approved EUR obligation / labelled PDF / EPC QR / sandbox ledger.

## PRODUCTION INTEGRATION

External identity, independently managed ERP authority, disposable workers and an idempotent bank adapter.